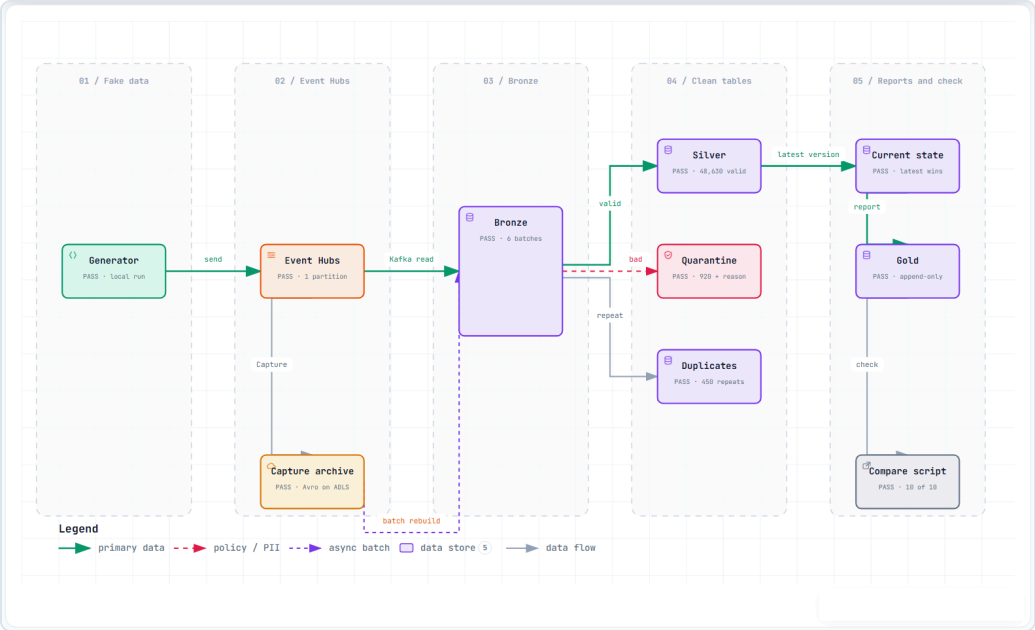
Trade-event pipeline on Azure

A learning project with fake data, built in about a week by one person. It processes messy trade events and checks its own output against an answer written before the run. Why it matters: in trading, a record counted twice, or arriving late, gives wrong numbers.

RESULT (FULL RUN, 5 NOVEMBER 2026)

50,000 fake events, Event Hubs to Databricks	**All processed. 0 lost.**
Output against the known answer	**10 of 10 checks equal**
Rebuild from the saved archive	**6 of 6 tables equal. Damaged tables can be rebuilt with the same result.**
Restart after a pause	**Resumed at the right place. No event twice, none missing.**
Tests	**185 tests, passing on Ubuntu in GitHub Actions**
Cost	**About 7 USD for the two Azure runs (including two hung attempts, not separated), about 10 USD in all. Everything deleted after each run.**

HOW IT FLOWS



THE PROBLEM IT SOLVES

- **Duplicates:** counted once, repeats listed (450).
- **Late and out-of-order events:** newest version wins; late events flagged, never dropped (843).
- **Bad records:** kept in quarantine with a reason (920).
- **Reports that change:** numbered versions. Old ones are never edited.

IN PLAIN WORDS

Bronze = raw events. Silver = cleaned events. Gold = reports. Quarantine = bad records, with a reason. Capture = an automatic copy of every event.

WHAT IS NOT PROVEN

- **The test is self-made:** one generator makes the events, the faults and the answer key. It shows the pipeline does what the design says, not that the design fits real feeds.
- All data is fake. Not production code. No compliance claim.
- One lane of the stream (one partition), one run. No test that an unauthorized user is blocked.
- Small batches, not real time.

BUILT WITH

Azure Event Hubs (Kafka, Capture), Azure Databricks serverless, Unity Catalog, Delta Lake, PySpark streaming, PowerShell, pytest, GitHub Actions.

Code, results and every claim with its evidence: https://github.com/sr-dotcom/trade-event-pipeline · Interactive walkthrough: https://sr-dotcom.github.io/trade-event-pipeline/